

Introduction

Tomatoes are among the most widely cultivated and commercially vital crops, making the study of their ripening process crucial for reducing food waste and extending shelf life¹. Traditionally, the *CNR* (colorless non-ripening) gene was seen as the main regulator of ripening, as its methylation of the 270bp region of the promoter region in *cnr* mutants delayed ripening, producing mealy textured firmer fruit that lack color¹. Specifically, this region is 2.4kb upstream of the coding sequence.⁴

This mutation allows farmers to delay ripening until reaching stores, where exposure to ethylene gas can trigger ripening, reducing waste for retailers and extending freshness for consumers.¹

However, recent findings show that removing this *CNR* region doesn't entirely halt ripening.⁵ Large deletions in the *Cnr* region of the gene produced orange or red-ripe fruits, suggesting other genes may also regulate the process.⁵ This study attempts to explore and identify those additional genes involved in tomato ripening and their location on the genome. Additionally, we aim to study the differences in expression of these genes in comparison to wild-type plants to determine which other genes on the pathway ultimately determine the tomatoes phenotypes.

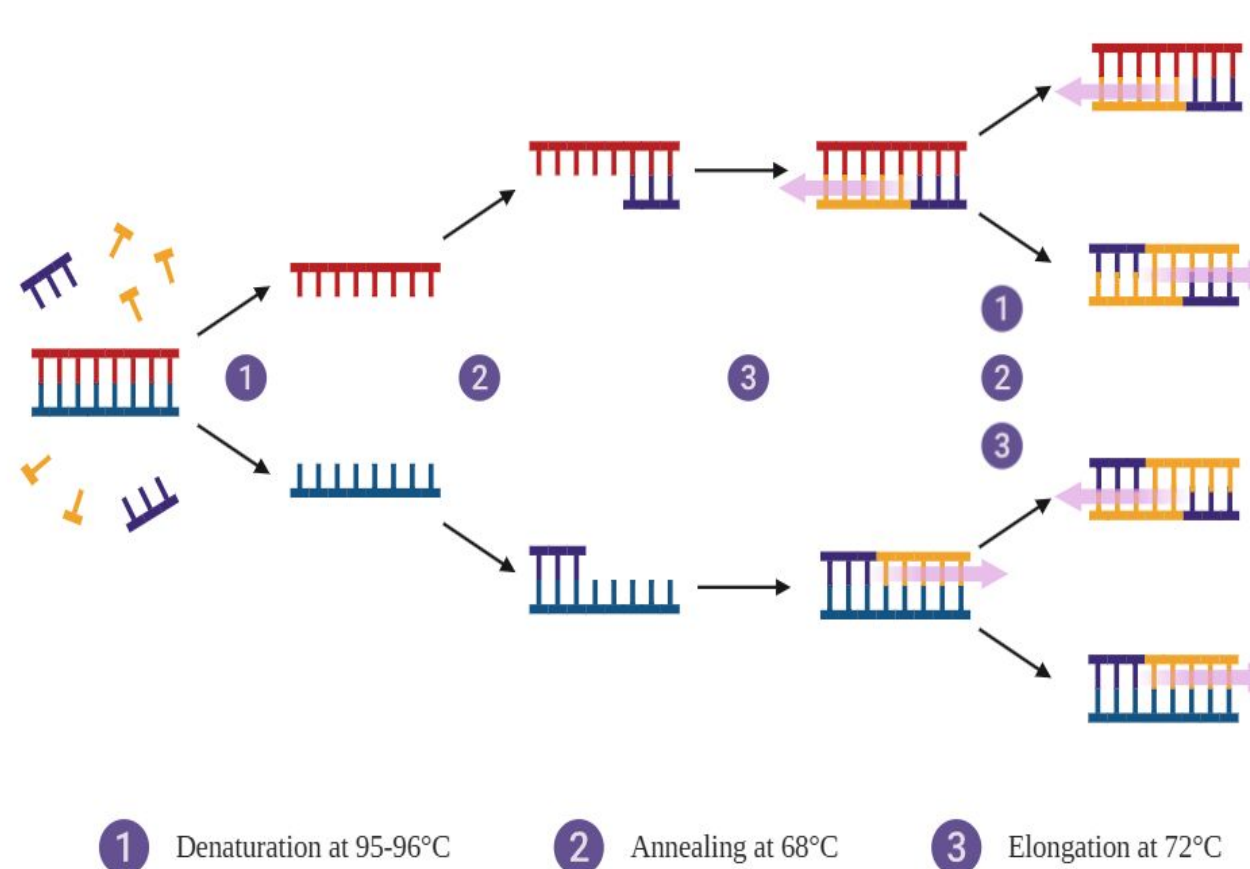
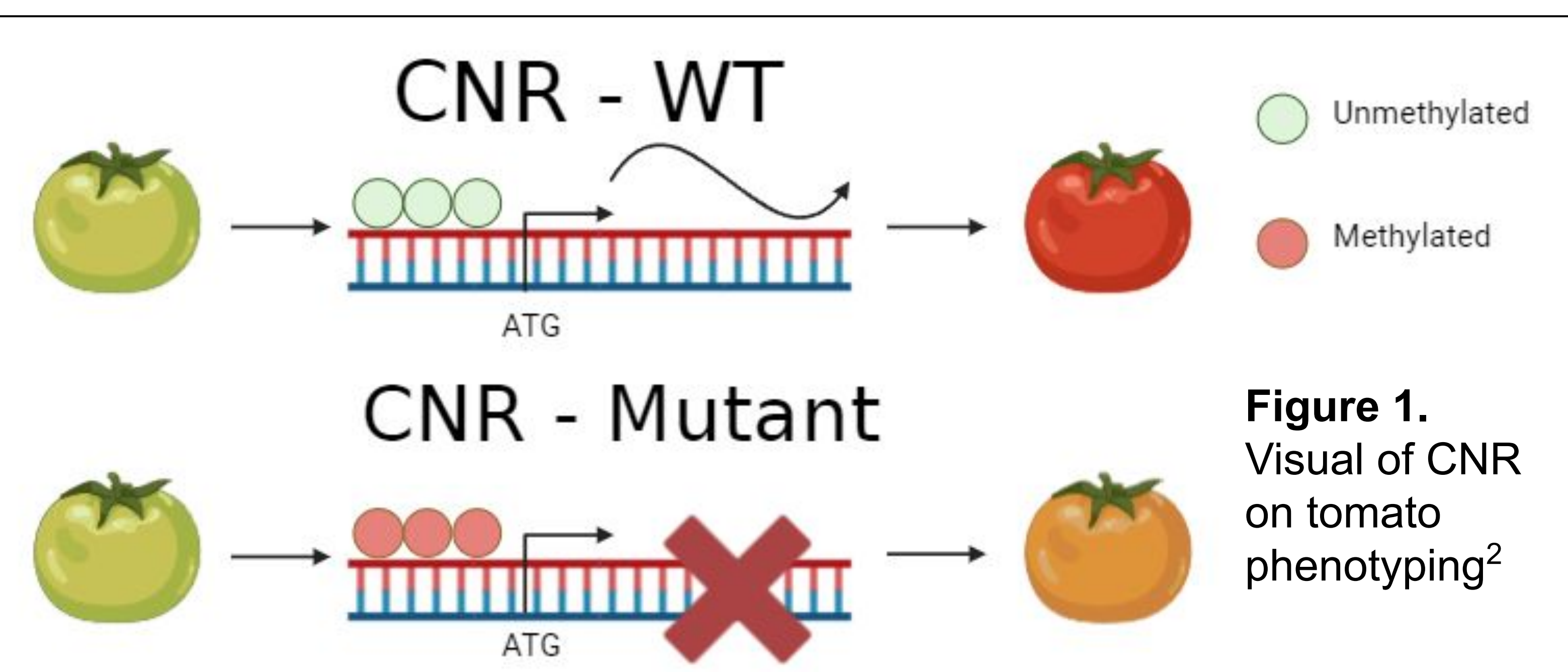


Figure 2. Visual of PCR reaction

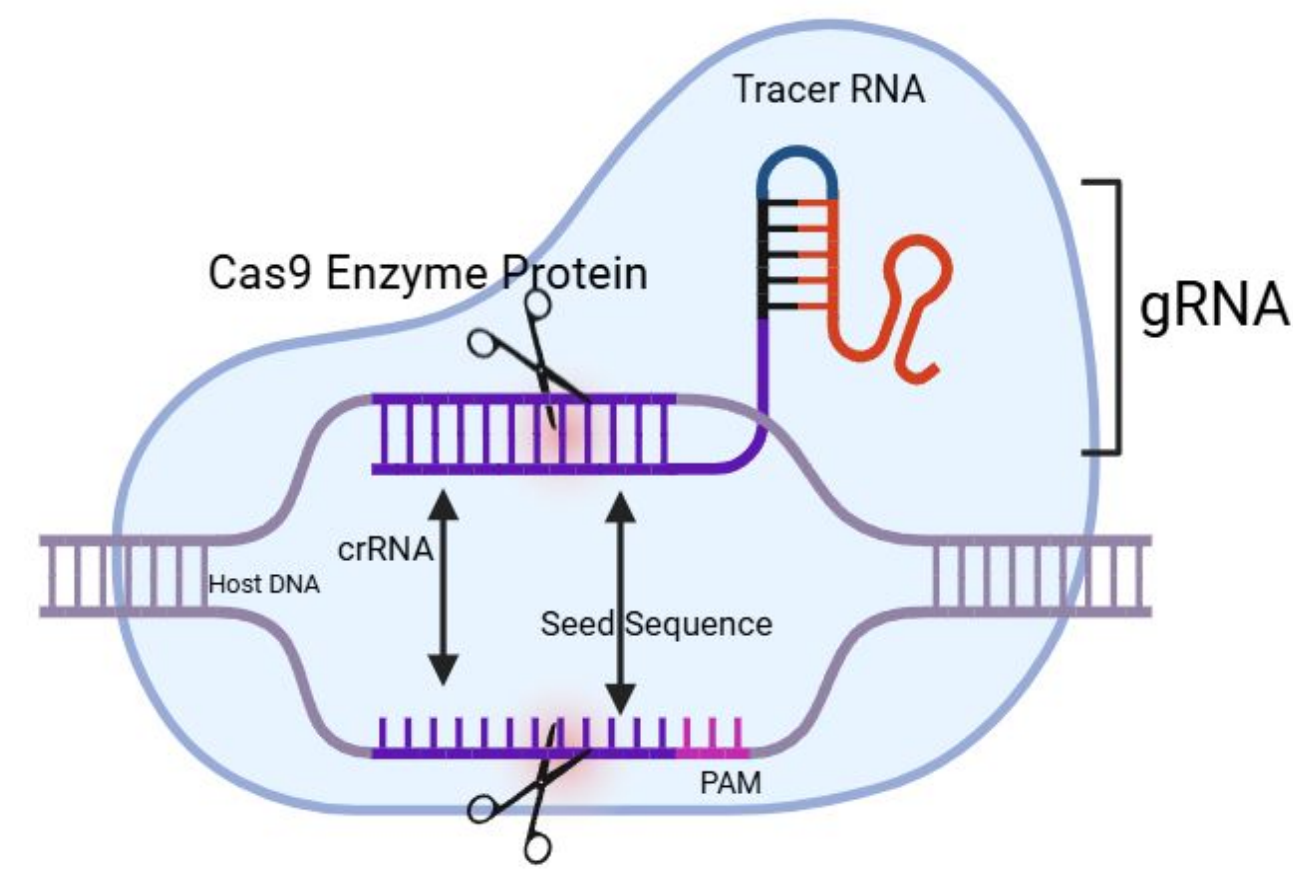


Figure 3. Visual of Cas9 Enzyme

Materials and Methods

1. Phenotyping:

Fruits were observed and compared for physical differences between the CRISPR-edited plants and wild-type.

2. DNA Extraction:

Plants were sampled, and DNA was extracted from ground leaf tissue. The process included cell lysis, DNA binding, washing, and elution.

3. CRISPR-Cas9 Targeting:

The Cas9 enzyme was employed through CRISPR to introduce precise edits in the *CNR* gene region. It is important to note that residual Cas9 during growth can lead to unintended changes in the DNA sequence.

4. Polymerase Chain Reaction (PCR):

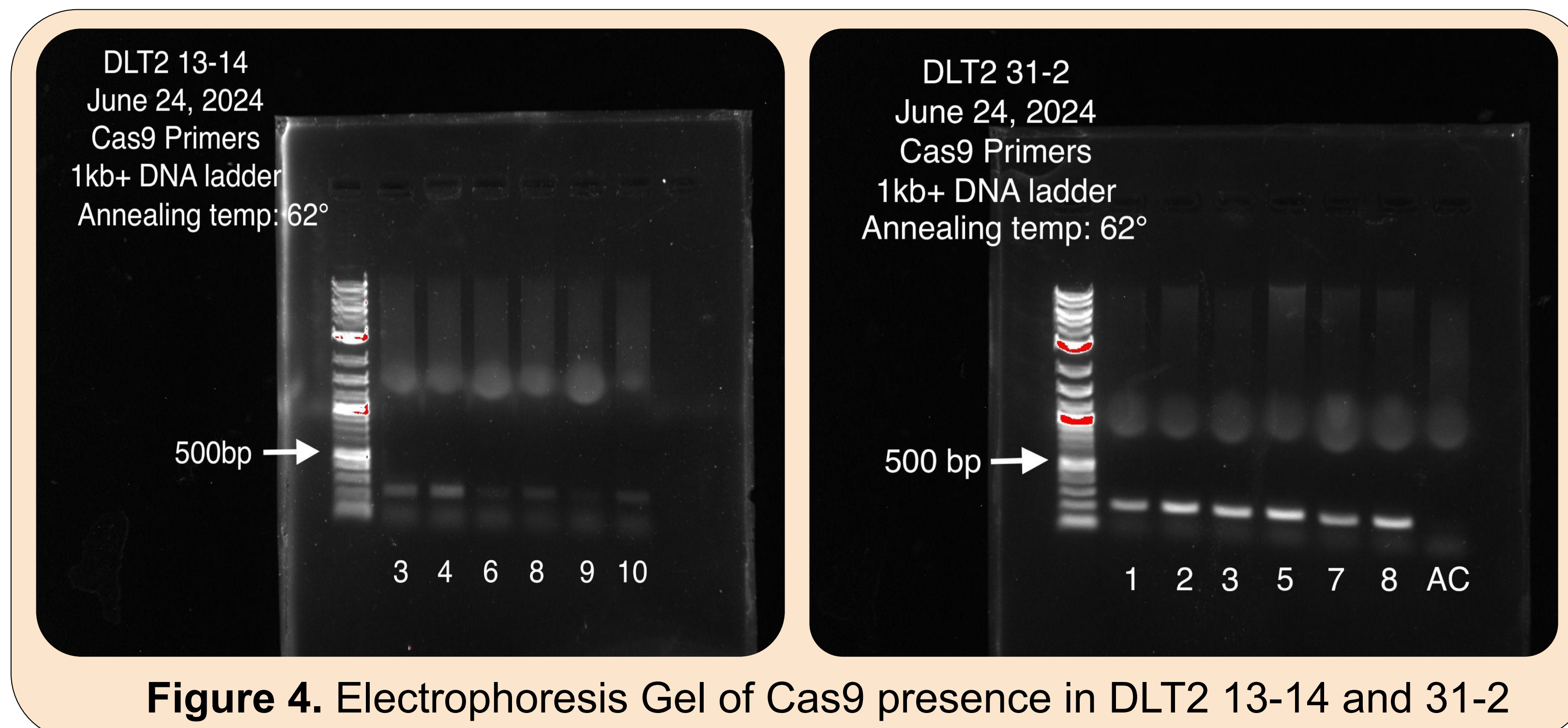
Primers were designed to amplify the promoter region of the *CNR* gene. PCR amplification confirmed DNA segments with targeted deletions introduced by Cas9.

5. Analysis:

After PCR, the amplified DNA was sequenced to determine the precise locations of deletions in the genome.

6. RNA Sequencing (RNA-Seq):

RNA-Seq was conducted to assess gene expression differences between the *cnr* mutants lines and wild-type plants. Transcription factors and genes that were up- or down-regulated in mutants were identified.

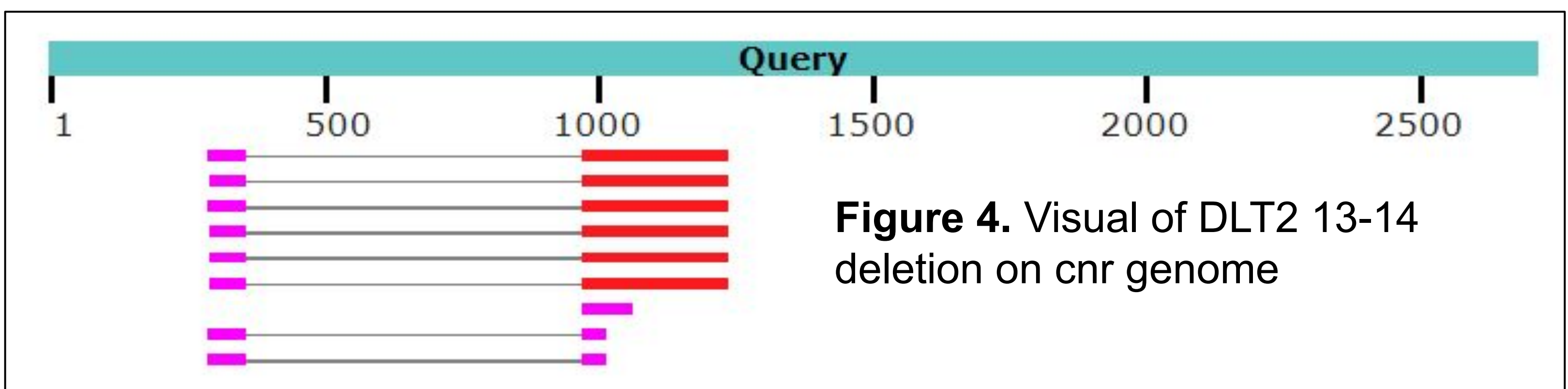


Results

- **DLT2 13-14 Line:** Showed minimal Cas9 presence; ripened to red at BK+25.
- **31-2 Line:** Exhibited Cas9 activity; ripened to orange at the same stage.
- **Hypothesis:** Cas9 activity in 31-2 line likely impacted ripening.
- **Observation:** 600 bp deletion in DLT2 13-14 removed the methylated *CNR* region, but partial ripening still occurred, suggesting compensatory mechanisms in ripening.



Figure 5. DLT2 31-2 vs 13-14 at BK+25



Conclusions and Further Directions

Our results suggest that the *cnr* gene is not the sole master regulator of ripening but rather a gain of function mutation, as its deletion did not result in a complete lack of ripening. This indicates that ripening is likely controlled by a network of multiple genes rather than a single gene. The presence of Cas9 in some plant lines may have introduced unintended genomic changes, potentially inhibiting ripening and complicating our analysis. Additionally, we identified the annexin p35 gene through the large up-regulation in the *cnr* mutant compared to the WT. We observed a difference in expression on average of over 35 times. Annexin P35 is known to be heavily involved in calcium dependent phospholipid binding and calcium ion binding which stabilizes membrane structures and organizes membrane domains³. This could explain the firmer more mealy texture of the *cnr* mutant. For future studies, conducting RNA sequencing on our specific tomato lines, would provide deeper insights into gene expression differences. Additionally, measuring ethanol production could offer further understanding of the metabolic differences between the lines.

References

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